

BUS 123 - Solving Business Problems with Technology

EXCEL-M01-L01

BUS 123 — EXCEL-M01-L01 — Creating and Editing Worksheets

PRE-READING

BUS 123 — EXCEL-M01-L01 — Creating and Editing Worksheets

BUS 123 · EXCEL-M01-L01 · CREATING AND EDITING WORKSHEETS

Course: Solving Business Problems with Technology · Fall 2026 · **Track:** EXCEL · **Module:** M01 · **Lesson:** L01

Case Study Company: Harborside Medical Center

1 · CONNECT TO PRIOR KNOWLEDGE

In the course introduction, you created a BUS 123 folder in OneDrive and saved your first file. That habit—**save first, save often, and save to the right place**—now moves inside Excel.

An Excel **workbook** is the file you save. A workbook can contain several **worksheets**, shown as tabs along the bottom. Each worksheet contains cells arranged in rows and columns. Business analysts use those cells to store assumptions, construct formulas, and show evidence for a decision.

Harborside Medical Center will use aggregate department and payer data in this lesson. The data are simplified for training and contain no patient-level information.

Your goal before class

Be ready to locate a cell, explain what is stored in it, distinguish a displayed format from an underlying value, and predict what should happen when a formula is copied.

2 · FIND YOUR WAY AROUND EXCEL

The five interface zones

Zone	Where it appears	What it tells you or lets you do
Ribbon	Across the top	Organizes commands into tabs such as Home, Insert, Formulas, and Data.
Name Box	Left of the Formula Bar	Shows the address of the active cell, such as C8 ; you can also type an address to jump there.
Formula Bar	Above the worksheet grid	Shows the underlying value or formula stored in the active cell.
Worksheet grid	Main work area	Organizes cells by column letter and row number.
Sheet Tabs	Along the bottom	Move among worksheets inside the same workbook.

The intersection of a column and row is a **cell**. Column C and row 8 identify cell `C8`. A colon means “through,” so `C8:C11` means every cell from C8 through C11.

Navigation shortcuts

Excel commands vary slightly by platform. Formula syntax and cell addresses remain the same.

Goal	Windows	Mac
Return to cell A1	<code>Ctrl+Home</code>	<code>Control+Home</code> or <code>Control+Fn+Left Arrow</code> on some MacBooks
Move to the last used cell	<code>Ctrl+End</code>	<code>Control+End</code> or <code>Control+Fn+Right Arrow</code> on some MacBooks
Move to the edge of the current data region	<code>Ctrl+Arrow</code>	<code>Command+Arrow</code>

Use both the Name Box and Formula Bar

The Name Box answers “**Where am I?**” The Formula Bar answers “**What is actually stored here?**” A displayed result alone cannot answer both questions.

3 · SEPARATE THE STORED VALUE FROM ITS DISPLAY

Formatting changes how a value appears; it does not change the stored value.

Stored value	Display format	What the worksheet may show
84200	Currency or Accounting	\$84,200
0.62	Percentage	62%
0.41	Percentage	41%

If you click a cell that displays 62%, the Formula Bar may still show 0.62. The displayed percentage is communication; the stored decimal is what formulas use.

Useful formatting decisions

- Format dollar amounts as Currency or Accounting.
- Format rates and shares as Percentage.
- Use clear headers and enough column width to show complete labels.
- Let Excel's default alignment help distinguish text from numbers, but treat alignment as a formatting convention—not a universal rule.
- Use borders and fills to clarify sections rather than decorating every cell.

⚠ Common mistake: entering 62 instead of 62%

62% and 0.62 represent the same stored rate. Entering 62 and applying Percentage format produces 6200%, which is a different value.

4 · BUILD FORMULAS WITH CELL REFERENCES

Every Excel formula begins with an equals sign. A cell reference tells Excel where to find an input.

Relative references move

A relative reference such as C17 changes when a formula is copied to another row. If the formula in C23 is copied down one row, C23 becomes C24.

Absolute references stay fixed

An absolute reference such as \$C\$21 keeps both the column and row fixed. Use it when every copied formula must use the same shared assumption.

Harborside's starter workbook uses a **simplified training contract**:

Expected reimbursement = stated charge × supplied reimbursement rate

The supplied rate is stored in cell C21 on the **Charge Analysis** tab. The first reimbursement formula is:

Excel syntax: =C23*\$C\$21

Manual check: \$84,200 × 62% = \$52,204

When copied from D23 to D24:

- C23 changes to C24 because the department charge should move with the row.
- \$C\$21 remains \$C\$21 because every department uses the same supplied rate.

On Windows, F4 cycles a reference through its locking modes. On Mac, use Command+T or F4; MacBook function-key settings may affect whether you also need the Fn key. The important decision comes first: **identify what must move and what must remain fixed.**

Model boundary

This charge-times-rate calculation is a simplified classroom assumption supplied for this lesson. Real payer contracts and Medicare payment methodologies vary. Do not describe this training calculation as a universal hospital reimbursement rule.

5 · CHOOSE A FORMULA FROM THE BUSINESS QUESTION

Do not choose a function merely because you remember its name. First identify the evidence the manager needs.

Business question	Evidence needed	Excel pattern
What are total monthly charges?	One combined total	=SUM(C6:C10)
What is the typical department charge?	Arithmetic mean	=AVERAGE(C6:C10)
How many numeric charge entries are present?	Count of numeric cells	=COUNT(C6:C10)
What is the largest charge?	Highest value	=MAX(C6:C10)
What is the smallest charge?	Lowest value	=MIN(C6:C10)

Verification is part of the model

A worksheet without visible errors can still be wrong. Use evidence that tests the model's logic:

1. **Formula Bar check:** confirm the cell contains a formula, not a typed answer.
2. **Range check:** confirm the formula includes every intended input and no unintended cells.
3. **Reference check:** confirm shared assumptions stay fixed when formulas are copied.
4. **Reconciliation check:** calculate the same total a second way and subtract. A correct difference should equal zero.
5. **Reasonableness check:** ask whether the result fits the business context.

For example, the five Harborside department reimbursements should total the same amount as total charges multiplied by the shared rate. If those two totals differ, the model does not reconcile.

⚠ Syntax errors and logic errors are different

Excel often flags a syntax error. It may not flag a formula that uses the wrong range, omits a row, or hard-codes an assumption. Those silent logic errors require an audit check.

6 · PREPARE FOR CLASS

Answer these five questions before class. Bring your reasoning, not only a final answer.

- [Retrieval]** Which interface element shows the address of the active cell, and which element shows the value or formula stored in that cell?
- [Interpretation]** A cell displays 62% , while the Formula Bar shows 0.62 . Did formatting change the stored value? Explain.
- [Reference prediction]** Cell D23 contains =C23*\$C\$21 . What should the formula become when copied to D24? Identify what moves and what stays fixed.
- [Method choice]** A manager asks, “What is the highest department charge?” Which Excel pattern matches the evidence needed, and why would SUM answer a different question?
- [Error diagnosis]** A student types 176886 into a total cell and sees no formula error. Why is that not convincing proof that the reimbursement model is correct? Name one stronger check.

KEY VOCABULARY

Term	Meaning
Workbook	The Excel file that can contain one or more worksheets.
Worksheet	One tabbed grid inside a workbook.
Active cell	The currently selected cell.
Cell address	A column letter and row number, such as C8 .
Range	A group of cells, such as C8:C11 .
Formula Bar	The interface area that shows the active cell's underlying value or formula.
Relative reference	A reference that changes when copied.
Absolute reference	A reference, such as \$C\$21 , that remains fixed when copied.
Reconciliation	An independent comparison used to prove that two logically equivalent totals agree.